

Physics-Informed AI-Assisted TCAD Design

Device-Level Power-Performance-Area (PPA) Optimization
of Gate-All-Around (GAA) Nanosheet Transistors

M.Tech AI Project — Overview Document

Domain	Semiconductor Device Physics + Machine Learning
Target Device	Gate-All-Around (GAA) Nanosheet FET
Simulation Tool	Sentaurus TCAD
Optimization Goal	Power, Performance, Area (PPA)
ML Approach	GNN Surrogate + RL / NSGA-II Optimization

1. What Is This Project About?

As transistors shrink below 5 nanometers, the industry is moving from FinFET to **Gate-All-Around (GAA) Nanosheet FET** architecture. Samsung's MBCFET and Intel's RibbonFET are real-world examples. These transistors use stacked horizontal nanosheets surrounded by the gate on all four sides, giving better electrostatic control.

Designing these transistors involves choosing the right geometry — sheet width, thickness, number of sheets, gate length, and more. Each combination produces different electrical behavior. Running a single TCAD simulation takes minutes to hours. Exploring thousands of combinations this way is impractical.

Our idea: Run a small set of TCAD simulations (~200–250), train an AI model to predict transistor performance from geometry parameters, and then use that AI model to rapidly explore millions of design combinations to find the optimal ones.

2. The Core Idea — In Simple Terms

1. **Step 1 — Build:** Create a reference GAA nanosheet transistor in Sentaurus TCAD software.
2. **Step 2 — Vary:** Systematically change the geometry parameters (width, thickness, length, etc.) to create ~200–250 different transistor designs using Design of Experiments (DOE).
3. **Step 3 — Simulate:** Run TCAD simulation on each design to get electrical outputs (current, leakage, threshold voltage, etc.). This is our ground truth data.
4. **Step 4 — Train AI:** Train a machine learning model that learns: *"Given these geometry parameters, here are the electrical characteristics."*
5. **Step 5 — Optimize:** Use an optimization algorithm (NSGA-II or Reinforcement Learning) that queries the AI model millions of times to find the best trade-offs between power, performance, and area.
6. **Step 6 — Validate:** Pick the top optimized designs and re-run them in TCAD to confirm the AI's predictions are accurate.

3. Device Parameters We Are Optimizing

These are the **input geometry parameters** — the design knobs we can turn to change how the transistor behaves:

Parameter	Symbol	Range	What It Means
Gate Length	L_g	10 – 16 nm	Length of the gate electrode along the channel. Controls short-channel effects.
Nanosheet Width	W_{NS}	15 – 35 nm	Width of each horizontal nanosheet. Wider = more drive current.
Nanosheet Thickness	T_{NS}	4 – 7 nm	Thickness of each nanosheet. Thinner = better electrostatic control.

Parameter	Symbol	Range	What It Means
Spacer Length	L_{sp}	3 – 7 nm	Length of the gate dielectric spacer between gate and source/drain.
Sheet Spacing	T_{sp}	8 – 12 nm	Vertical gap between stacked nanosheets. Affects parasitic capacitance.
Number of Sheets	N_{NS}	2, 3, or 4	How many nanosheets are stacked. More sheets = more current drive.
Equiv. Oxide Thickness	EOT	0.7 – 1.3 nm	Effective gate oxide thickness. Thinner = stronger gate control.
Supply Voltage	V_{DD}	0.5 – 0.7 V	Operating voltage. Lower = less power but slower switching.

Table 1: Input design parameters for GAA nanosheet FET optimization.

4. What We Are Predicting — Output Metrics

These are the **electrical characteristics** the AI model learns to predict from the input parameters. They define the Power-Performance-Area (PPA) quality:

4.1 Power Metrics

- **I_{OFF} (Leakage Current):** Current flowing when transistor is OFF. Lower is better — determines static power consumption.
- **V_{DD} (Supply Voltage):** Directly scales dynamic power ($P \propto V^2$). Lower voltage = less power but potentially slower.

4.2 Performance Metrics

- **I_{ON} (Drive Current):** Current flowing when transistor is ON. Higher is better — determines switching speed.
- **V_{TH} (Threshold Voltage):** Minimum gate voltage to turn ON. Affects both speed and power.
- **SS (Subthreshold Swing):** How sharply the transistor switches from OFF to ON. Lower (closer to 60 mV/dec) = better switching.
- **DIBL (Drain-Induced Barrier Lowering):** Measure of short-channel effects. Lower is better — indicates good electrostatic control.

4.3 Area Metrics

- **C_{gg} (Gate Capacitance):** Total gate capacitance. Lower = faster switching and less dynamic power. Also relates to the physical footprint.
- **Footprint:** Determined by gate length $\times$ nanosheet width $\times$ number of sheets. Smaller area = more transistors per chip.

Metric	Symbol	PPA Category	Goal
Drive Current	I_{ON}	Performance	Maximize
Leakage Current	I_{OFF}	Power	Minimize
Threshold Voltage	V_{TH}	Performance + Power	Target range
Subthreshold Swing	SS	Performance	Minimize
DIBL	DIBL	Performance	Minimize
Gate Capacitance	C_{gg}	Area + Power	Minimize

Table 2: Output metrics — what the AI model predicts and the optimization goals.

5. The Fundamental Trade-Off

These metrics **conflict** with each other — improving one often worsens another:

If You Improve...	Then This Gets Worse...	Why?
Higher I_{ON} (faster)	Higher I_{OFF} (more leakage)	Thinner oxide / shorter gate increases both on and off current
Lower V_{DD} (less power)	Lower I_{ON} (slower)	Less voltage = less drive current
More sheets (more current)	Larger area + more capacitance	Stacking increases height and C_{gg}
Shorter gate (smaller)	Worse DIBL + SS	Short-channel effects worsen

Table 3: Key trade-offs in GAA nanosheet FET design.

This is why we need **multi-objective optimization** — there is no single "best" design. Instead, we find a set of **Pareto-optimal** designs where no metric can be improved without worsening another. The designer then picks from this set based on application needs.

6. Machine Learning Models

We plan to train and compare multiple surrogate models:

Model	Type	Role	Why
XGBoost	Tree Ensemble	Baseline surrogate	Best for small tabular data. $R^2 > 0.99$ in TCAD literature.
Random Forest	Tree Ensemble	Baseline surrogate	Robust, less overfitting. Supervisor recommended.
GNN (Graph Attention)	Deep Learning	Novel surrogate	Operates on TCAD mesh directly. Captures spatial physics.
Physics-Informed ANN	Deep Learning	Novel surrogate	Embeds Grove-Frohmman model as constraint. Less data needed.

Table 4: Machine learning models planned for surrogate training.

7. Optimization Approach

Once the surrogate model is trained, we use it to explore the design space and find optimal configurations:

Method	How It Works	Novelty
NSGA-II	Evolutionary algorithm. Maintains a population of designs, evolves them over generations to find Pareto front.	Standard baseline. Well-established.
Reinforcement Learning (PPO)	Agent learns a policy: given current parameters, which parameter to adjust next to maximize PPA reward.	Very novel for transistor design. Few have done this.

Table 5: Optimization methods for PPA design space exploration.

8. Complete Workflow



